Biomedical Signal Processing exam 17/12/2025

Implementation of “paper 4” [1]:

“A wavelet-based ECG delineation algorithm for 32-bit integer online processing” (2011), Luigi Y Di Marco and Lorenzo Chiari



Giacomo Sacco 82389A

Outline



1. The problem
2. Datasets description
3. Algorithm description
4. Results
5. Conclusions

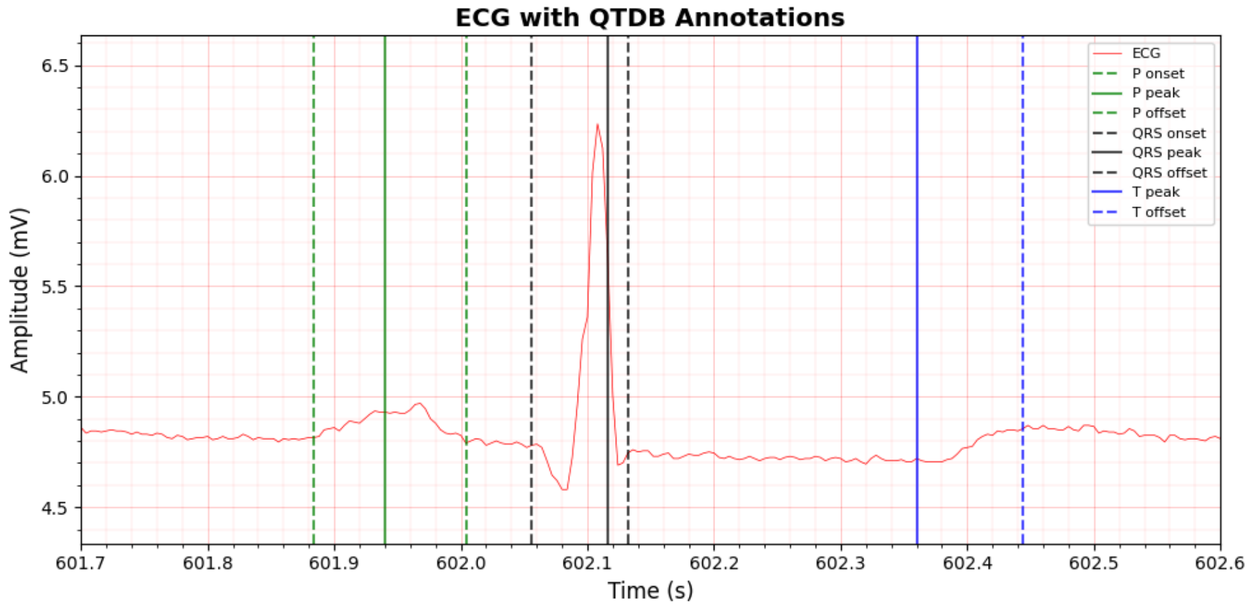
1 The problem to solve: **ECG delineation**

Core Objective

Build an online, low-complexity algorithm designed for devices with limited processing power

How?

Implement a wavelet transform method based delineation method that strictly use 32-bit integer linear algebra

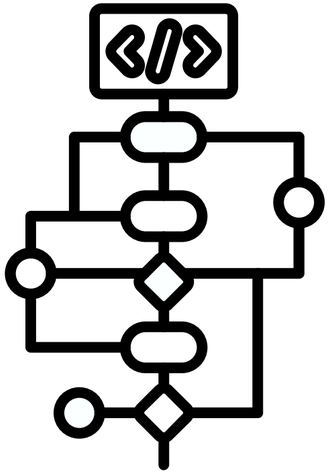


From QTDB recording “sel100”, q1c annotation

2 Datasets

	MIT-BIH Arrhythmia Database (MITDB)	European ST-T Database (EDB)	QT Database (QTDB)
Purpose	Validate QRS detection	Validate QRS detection	Validate P-QRS-T delineation
Content	≈ 24h ECG recordings	≈ 180h ECG recordings	105 15 min records drawn from various databases
Annotations	**109,010** annotated beats, focused on *arrhythmias*	**788,050** annotated beats, with various *abnormalities*	at least **30** beats per record
Pre-processing	Resampling to 250Hz	N/A	N/A

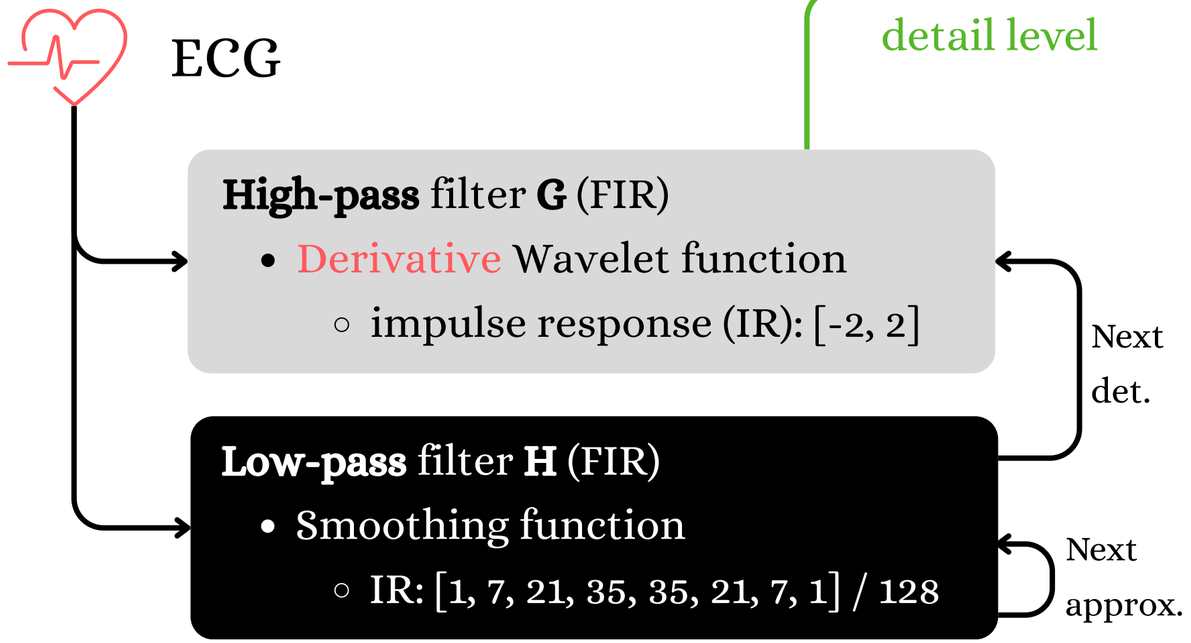
3 Algorithm steps



1. Wavelet Transform
2. QRS detection
3. QRS delineation
4. P-wave delineation
5. T-wave delineation

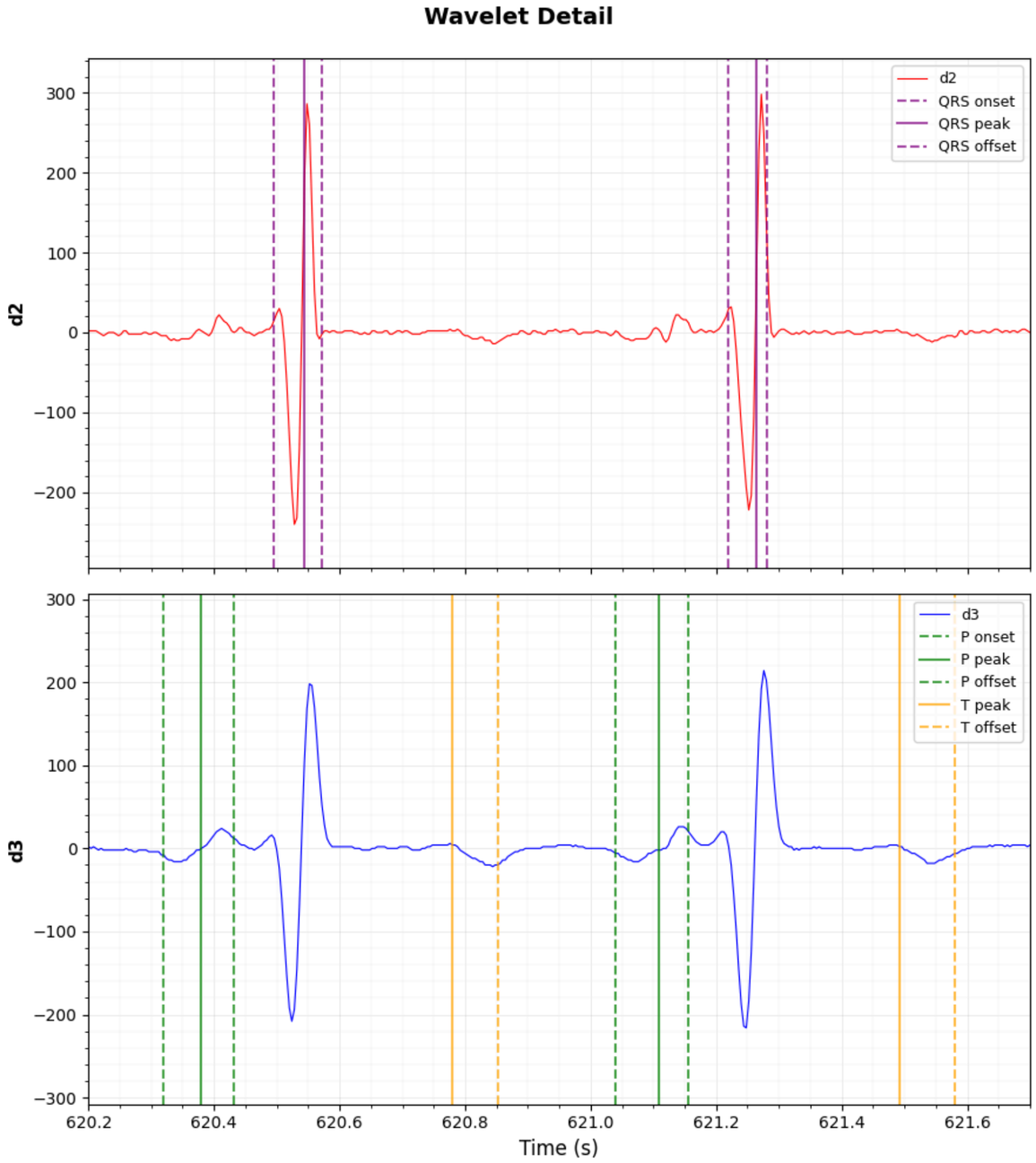
❸ Algorithm steps

1 Wavelet Transform



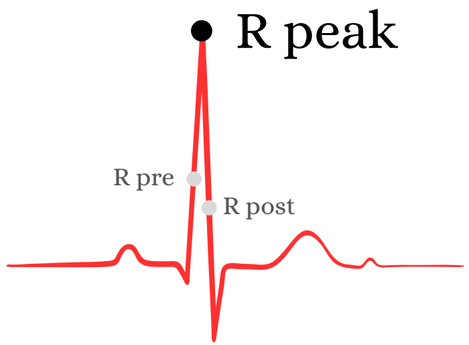
Undecimated DWT: add zeros in the filter every level to preserve original resolution

\+ **GROUP DELAY** adjustment



❸ Algorithm steps

2 QRS detection



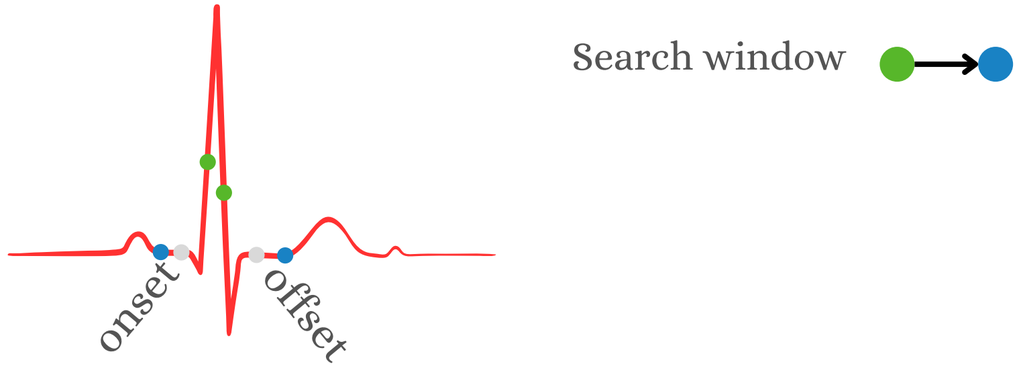
R pre/post = steepest slope before/after R peak

Search **R peaks**:

1. **Find** candidate R peaks using zero-crossings (ZCs) in *detail level 2*
2. **Validate** R peaks using **thresholds** on *detail levels 2 and 3*

$$\Delta d_n^2(n_Z) > \varepsilon_{QRS}^2 = \frac{11}{32} \cdot \frac{\sum_{n_Z} \Delta d_n^2(n_z)}{N}$$

3 QRS delineation

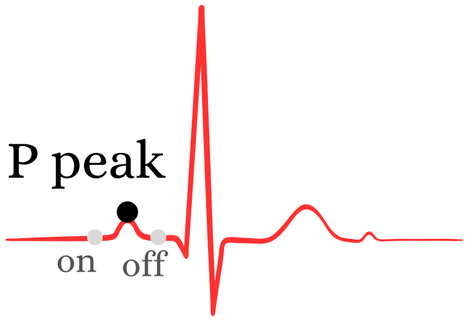


Search **QRS onset & offset**:

1. Start from R peak pre/post and go left/right
2. The first point that satisfy the **thresholds** is the onset/offset
3. Otherwise the **window end** (120 ms) is set as onset/offset

3 Algorithm steps

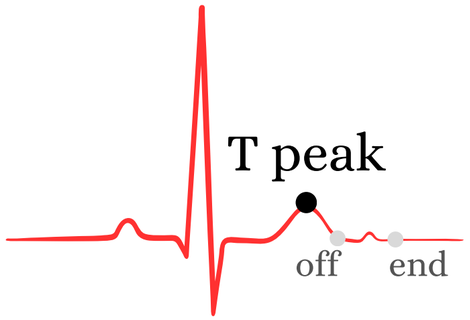
4 P wave delineation



Detail level 3 (lower frequency), **300 ms window** from QRS onset:

1. Search biggest peak using **ZCs**
2. Evaluate **bi-phasic** using a *threshold*
3. Define **P onset/offset** using two amplitude *thresholds*
4. If one element is not found within the window *discard the P delineation*

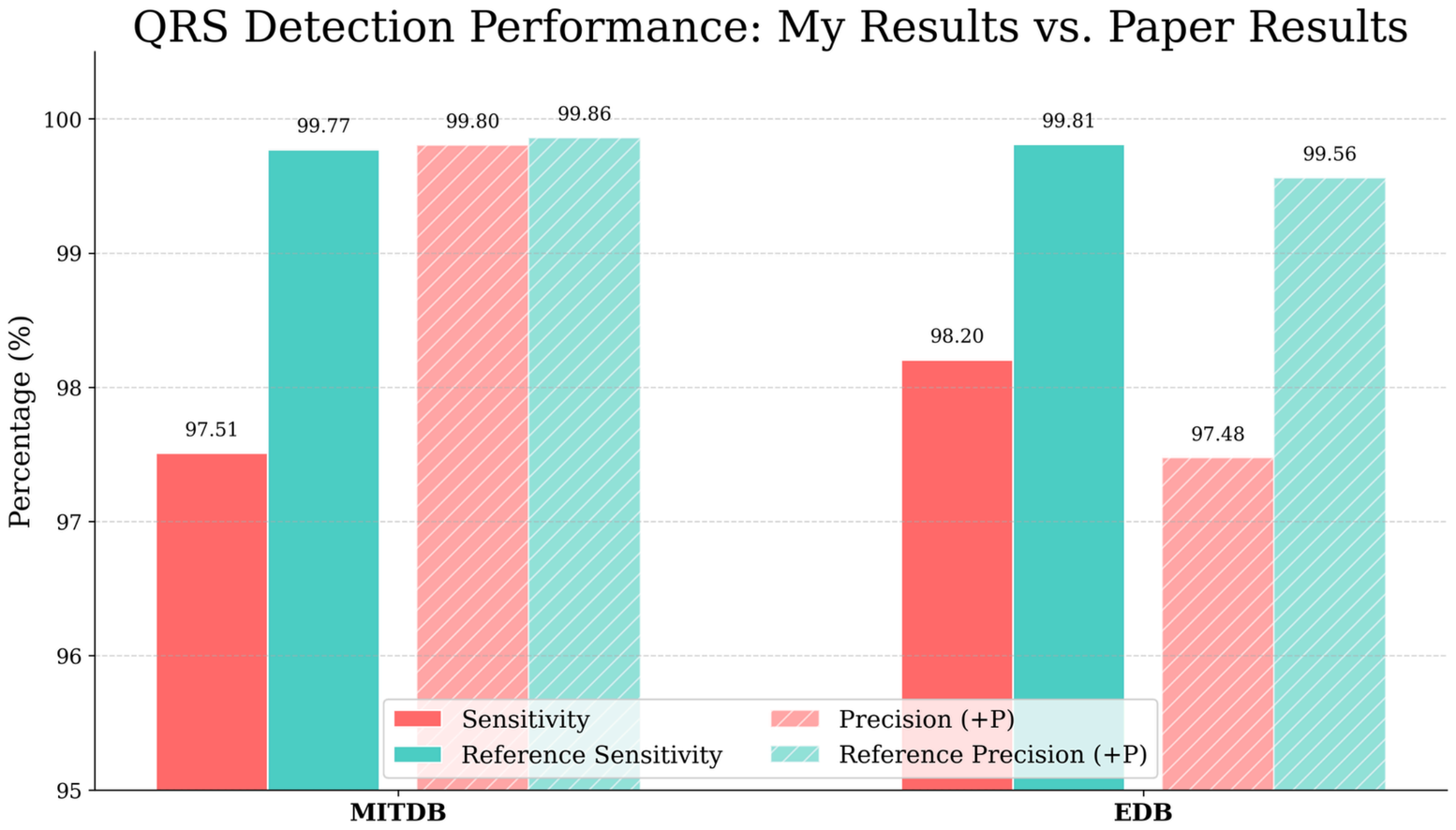
5 T wave delineation



Detail level 3 (lower frequency), window from QRS offset + 80ms to a point based on previous RR distance:

1. Search biggest peak using **ZCs**
2. Evaluate **bi-phasic** using two thresholds
3. Define *T offset* and *T end* using two amplitude thresholds

4 Results: QRS detection



Sensitivity:

The algorithm correctly identifies the majority of beats (97.51% on MITDB and 98.20 on EDB)

Precision:

The algorithm is accurate when detects a peak (99.80% on MITDB and 97.48% on EDB)

Clinical application:

This work sensitivity and precision would not be enough in a clinical setting (at least 99.5%)

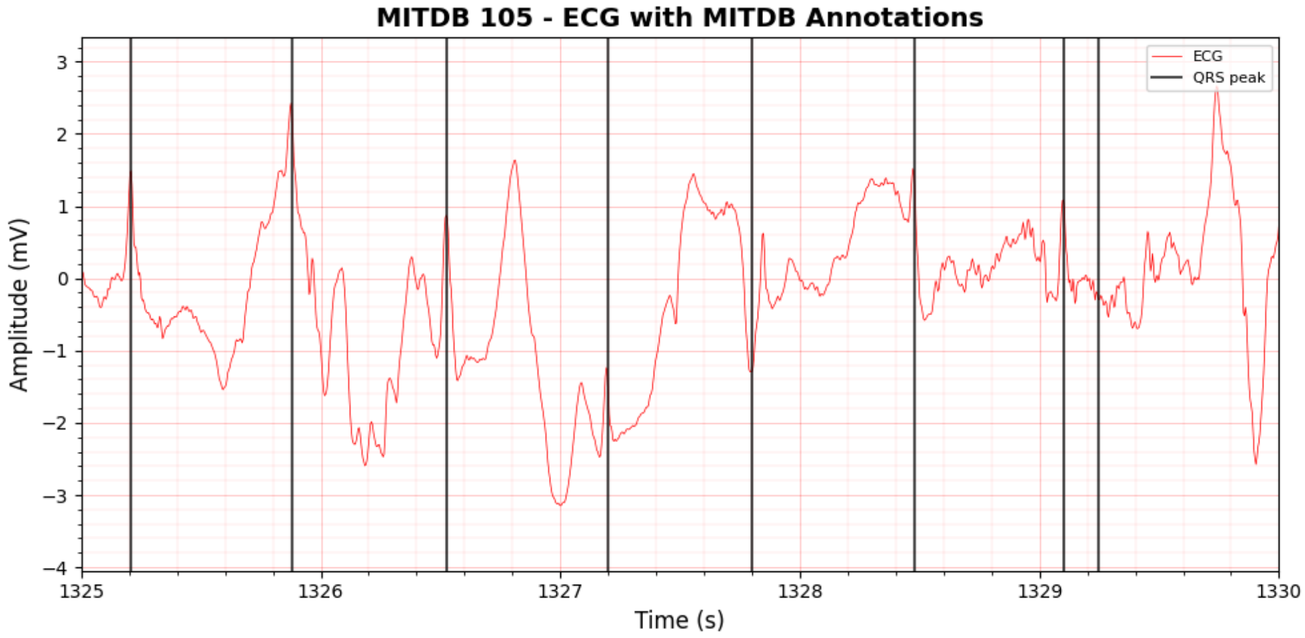
4 Results: QRS detection

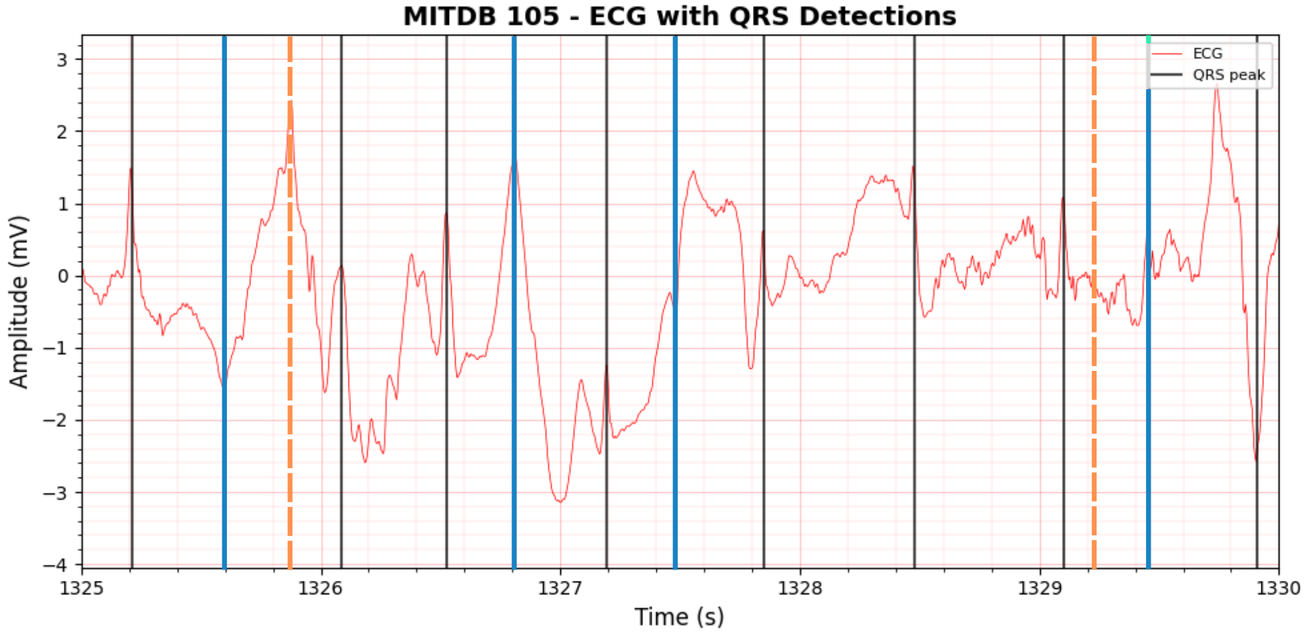
The images show how the algorithm struggle with **noisy signals**.
This problem is one of the principal causes of **97.51%** overall **sensitivity** of the algorithm on the MITDB.

Recording **105** from **MITDB**
Se: 95.45 %, +P: 97.74 %
Time range: 1325-1330 s

False Postitive

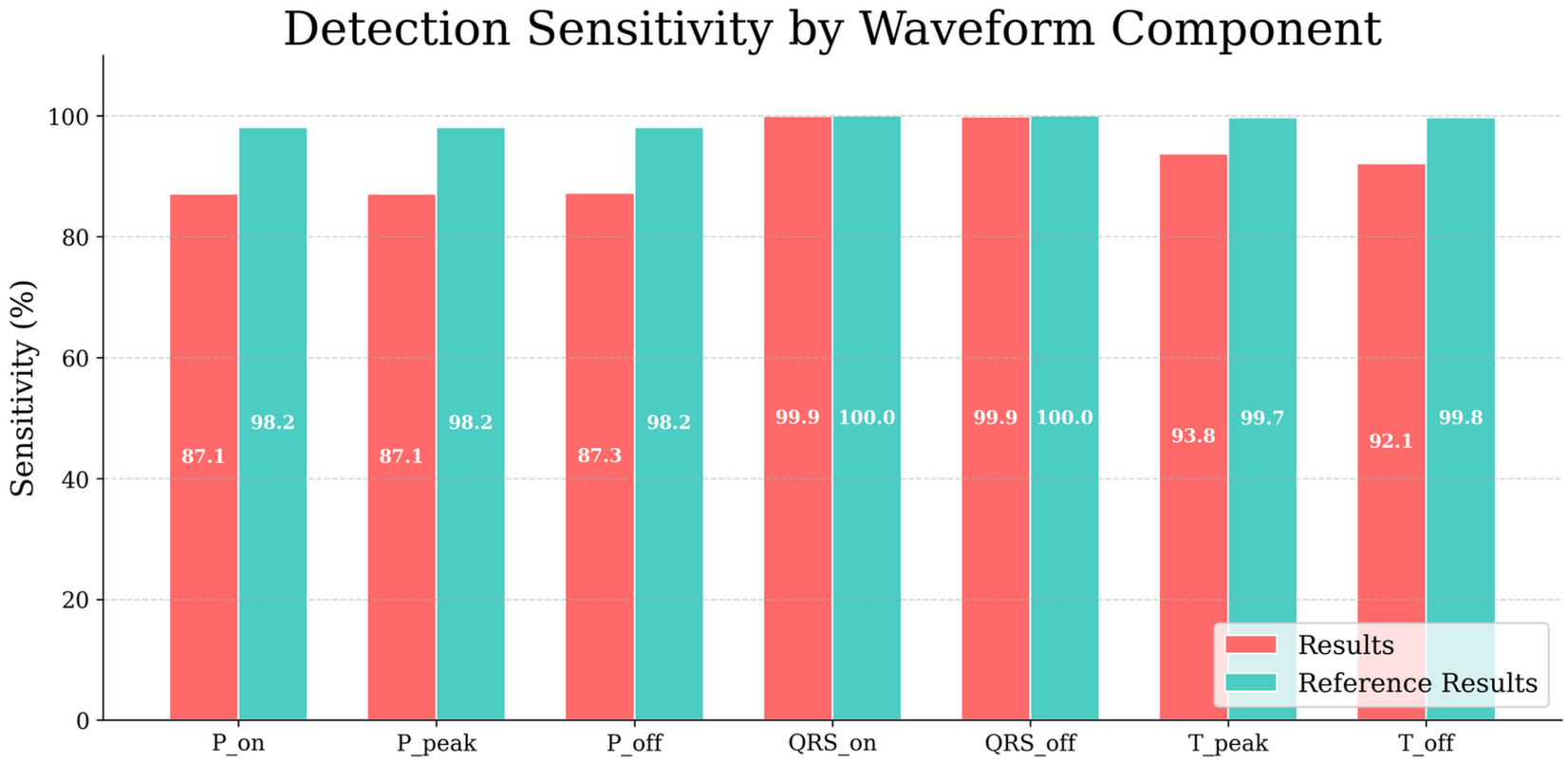
False Negative





4 Results: P-QRS-T Delineation

A component is accepted (TP) if within 150ms window (ANSI/AAMI-EC57:1998 standard) of the QTDB annotation



P wave

Most difficult part to correctly identify due to its low amplitude

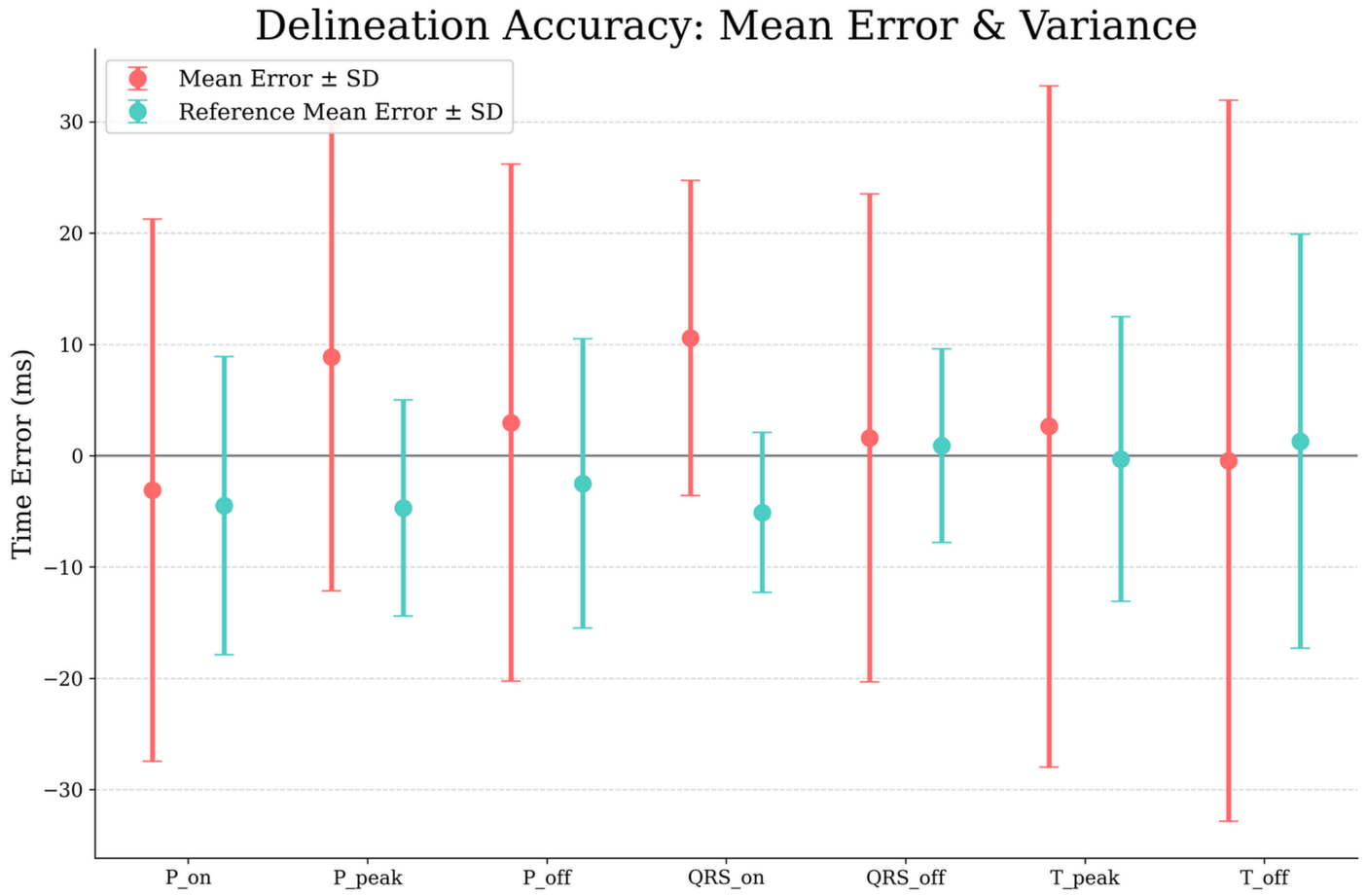
QRS delineation

Nearly perfect score

T wave

Gets better results than P wave thanks to its more prominent amplitude

4 Results: P-QRS-T Delineation



Mean error (ms)

Most of the points have a similar time error to the paper reference results. *P peak* and *QRS onset* are the ones with the most difference (≈ 10 ms)

Variance

The variance is around 2 times bigger than the paper reference results. Doesn't fit in the tolerance range defined by Common Standards for Electrocardiography (CSE) [2]

5 Conclusions

Implemented a the algorithm using strictly 32-bit integer **linear algebra** arithmetic

General performance is lower than the reference paper





Possible improvements:

- Improve algorithm robustness to noise
  - noise detection system
- Implement a T onset delineation function

Giacomo Sacco 82389A

Thank you!

giacomo.sacco@studenti.unimi.it

References

[1] Di Marco, L.Y., Chiari, L. A wavelet-based ECG delineation algorithm for 32-bit integer online processing. BioMed Eng OnLine 10, 23 (2011). https://doi.org/10.1186/1375-925X-10-23

[2] The CSE Working Party: Recomendations for measurement standards in quantitative electrocardiography. Eur Heart J 1985, 6:815-825